



Superstore Sales Data Analysis Using Python



Project Overview

This project performs **Exploratory Data Analysis (EDA)** on the Superstore dataset using Python. The main goal is to understand sales patterns, analyze product categories, examine sales distribution, and calculate the number of days taken to ship orders.

The project uses popular Python data science and visualization libraries such as **Pandas, NumPy, Matplotlib, and Seaborn**.



Objectives

The main objectives of this project are:

- To load and explore the Superstore dataset.
 - To understand the structure and summary statistics of the data.
 - To convert date columns into the proper datetime format.
 - To calculate the delivery time for each order.
 - To identify unique product categories.
 - To check for missing values in the dataset.
 - To analyze total sales for each category.
 - To visualize sales by category.
 - To understand the distribution of sales values.
-



Technologies and Libraries Used

- **Python**
 - **Pandas** – Data manipulation and analysis
 - **NumPy** – Numerical operations
 - **Matplotlib** – Data visualization
 - **Seaborn** – Statistical data visualization
 - **Google Colab / Jupyter Notebook** – Development environment
-



Dataset

The project uses the **Superstore Dataset**, which contains information related to orders and sales.

The dataset is loaded using:

```
df = pd.read_csv("/content/samplesuperstore.csv")
```

Important Columns Used

- `Order Date` – Date on which the order was placed.
- `Ship Date` – Date on which the order was shipped.
- `Category` – Product category.
- `Sales` – Sales amount.

Data Analysis Process

1. Importing Libraries

The required Python libraries are imported for data analysis and visualization.

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns
```

2. Loading the Dataset

The Superstore CSV dataset is loaded into a Pandas DataFrame.

```
df = pd.read_csv("/content/samplesuperstore.csv")
```

3. Exploring the Dataset

The first few rows of the dataset are displayed using:

```
df.head()
```

The structure and data types of the dataset are examined using:

```
df.info()
```

Summary statistics are obtained using:

```
df.describe()
```

4. Date Conversion

The `Order Date` and `Ship Date` columns are converted into datetime format.

```
df['Order Date'] = pd.to_datetime(df['Order Date'])  
df['Ship Date'] = pd.to_datetime(df['Ship Date'])
```

This allows date-based calculations and analysis.

5. Calculating Delivery Days

A new column named `Delivery Days` is created to calculate the number of days between order placement and shipping.

```
df['Delivery Days'] = (df['Ship Date'] - df['Order Date']).dt.days
```

This helps analyze the shipping time of orders.

6. Finding Unique Categories

The unique product categories available in the dataset are identified using:

```
df['Category'].unique()
```

7. Checking Missing Values

Missing values in each column are checked using:

```
df.isnull().sum()
```

This step helps determine whether data cleaning is required.

8. Category-wise Sales Analysis

The total sales for each product category are calculated using the `groupby()` function.

```
category_sales = df.groupby('Category')['Sales'].sum()
```

This provides a comparison of sales performance across different product categories.



Data Visualization

Sales by Category

A bar chart is created to visualize the total sales for each category.

```
category_sales.plot(kind='bar', figsize=(8,5))
plt.title("Sales by Category")
plt.ylabel("Total Sales")
plt.show()
```

This visualization makes it easier to compare the sales performance of different product categories.

Sales Distribution

A histogram is created using Seaborn to understand the distribution of sales values.

```
plt.figure(figsize=(8,5))
sns.histplot(df['Sales'], bins=30)
plt.title("Sales Distribution")
plt.show()
```

The histogram helps identify how frequently different sales values occur in the dataset.



Key Analysis Performed

The project performs the following analysis:

Analysis	Purpose
Dataset Exploration	Understand the dataset structure
Descriptive Statistics	Analyze numerical data
Date Conversion	Prepare dates for analysis
Delivery Days Calculation	Measure shipping time
Category Identification	Find available product categories
Missing Value Analysis	Check data quality

Analysis	Purpose
Category-wise Sales	Compare sales performance
Sales Distribution	Understand sales patterns

Insights

The analysis helps in understanding:

- Which product categories generate higher total sales.
- How sales values are distributed.
- The shipping duration of customer orders.
- The overall structure and quality of the dataset.
- Whether the dataset contains missing values.
- Different patterns and trends in Superstore sales data.

How to Run the Project

Step 1: Install Required Libraries

```
pip install pandas numpy matplotlib seaborn
```

Step 2: Download the Dataset

Place the `samplesuperstore.csv` file in the required project directory.

Step 3: Run the Python Code

Open the project in **Google Colab**, **Jupyter Notebook**, or any Python IDE and execute the code cells.

Step 4: View the Results

The program will display:

- Dataset preview
- Dataset information
- Descriptive statistics
- Unique categories
- Missing value counts
- Category-wise sales
- Sales by category bar chart
- Sales distribution histogram

Future Enhancements

The project can be further improved by adding:

- Profit analysis by category and region.
- Sales analysis by month and year.
- Top-selling products identification.
- Regional sales comparison.
- Customer segment analysis.
- Correlation analysis between sales and profit.
- Interactive dashboards using **Power BI** or **Tableau**.
- Predictive sales forecasting using Machine Learning.

Conclusion

This project demonstrates how Python can be used for **data preprocessing, exploratory data analysis, and visualization**. By analyzing the Superstore dataset, useful information about category-wise sales, sales distribution, and shipping duration can be obtained. The project provides a basic foundation for performing more advanced business and sales analytics.

Author

Sivarama Krishnan A

Project: Superstore Sales Data Analysis

Domain: Data Science and Data Analytics

Language: Python